

Reinforcement Learning-Guided Adaptive Temporal-Spectral Feature Fusion for Intelligent Heart Sound Analysis

Complete single-dataset implementation plan using the CirCor DigiScope phonocardiogram dataset

Prepared for SATYA KRISHNA V. | Final phase-wise methodology and exact download requirements

Recommended decision: Use only the CirCor DigiScope dataset v1.0.3 for the complete project. It provides PCG recordings from multiple auscultation locations, subject-level clinical metadata, murmur labels, and cardiac-cycle segmentation files. The current PhysioNet release reports 5,272 recordings from 1,568 subjects. [web:9]

1. Final research scope

The project will develop an RL agent that selects the most useful signal representation for each PCG segment. The selected configuration will include the wavelet family, decomposition level, frequency band, temporal-window size, and fusion strategy. The selected temporal, spectral, and wavelet features will then be fused using an attention-based network for murmur classification.

Primary task: binary murmur detection: murmur present versus murmur absent. **Secondary analysis:** normal versus abnormal outcome classification, used only after the primary pipeline is stable.

Why this dataset: CirCor includes waveform files, header files, segmentation files, and subject metadata, making it suitable for temporal-window selection and interpretability. The official challenge documentation identifies the four main data file types as WAV, HEA, TSV, and subject-level TXT files. [web:26]

2. Exact files to download

Download version 1.0.3 only. Do not mix files from versions 1.0.0, 1.0.1, or 1.0.2.

```
wget -r -N -c -np https://physionet.org/files/circor-heart-sound/1.0.3/
```

The complete release is available from PhysioNet; the same download command is documented by the PhysioNet Challenge page. [web:26]

File or folder	Download	Use in project
training_data/*.wav	Required	Raw PCG waveform for every auscultation location.
training_data/*.hea	Required	Sampling rate, sample count, and WFDB header metadata.
training_data/*.tsv	Required	S1, systole, S2, and diastole segmentation intervals.
training_data/*.txt	Required	Subject-level information, murmur label, outcome, and murmur descriptors.
training_data.csv	Required	Convenient metadata table and file mapping.
RECORDS	Recommended	List of available record identifiers.
SHA256SUMS.txt	Recommended	Integrity verification after download.
LICENSE.txt	Recommended	Dataset license and citation information.
README / database documentation	Recommended	Dataset definitions and file-format explanations.

The expected record pattern is:

```
subjectID_location.wav
subjectID_location.he
subjectID_location.ts
subjectID.txt
```

For example, a recording may have 49876_PV.wav, 49876_PV.he, and 49876_PV.ts, while 49876.txt contains subject-level information. The CirCor release contains 4,000-Hz recordings in its headers. [web:18]

3. Final directory structure

```
heart_sound_rl/  
  data/  
    raw/circor_v1.0.3/  
    interim/metadata/  
    interim/segments/  
    processed/  
  configs/  
  src/  
    data/  
    preprocessing/  
    features/  
    models/  
    rl/  
    training/  
    evaluation/  
    visualization/  
  experiments/  
  results/  
  notebooks/  
  requirements.txt  
  README.md
```

4. Phase-wise implementation

Phase 1 — Environment and download

Install Python, PyTorch, torchaudio, librosa, scipy, PyWavelets, pandas, scikit-learn, matplotlib, seaborn, and Gymnasium or Stable-Baselines3. Download CirCor v1.0.3 and verify the checksum file.

Phase 2 — Metadata construction

Read every WAV, HEA, TSV, and subject TXT file. Create `circor_records.csv` with `record_id`, `patient_id`, `location`, `paths`, `sampling rate`, `duration`, `murmur label`, `outcome label`, and `available murmur descriptors`.

Phase 3 — Patient-independent split

Split unique patient IDs into 70% train, 15% validation, and 15% test. Perform the split before making windows. Never place recordings from one patient in different splits.

Phase 4 — Quality control

Remove missing, clipped, silent, and unreadable signals. Store quality indicators rather than silently deleting every noisy signal, because robustness is part of the research question.

Phase 5 — Preprocessing

Resample from 4,000 Hz to 2,000 Hz for reduced computation, apply a broad 20–800-Hz bandpass filter, remove extreme amplitude values, and apply robust median/IQR normalization.

Phase 6 — Segmentation

Use TSV annotations for S1, systole, S2, and diastole. Generate overlapping candidate windows of 0.25, 0.50, 1.00, and 2.00 seconds. Keep the original timing information for interpretability.

Phase 7 — Candidate representations

Create candidate wavelets db4, db6, db8, sym4, and coif3; levels 3, 4, and 5; frequency bands 20–100, 100–200, 200–400, 400–800, and 20–800 Hz; and the four temporal-window sizes.

Phase 8 — Baseline models

Train raw-waveform CNN, fixed-STFT CNN, fixed-db6 wavelet CNN, and fixed temporal-spectral fusion. These models establish whether RL gives improvement.

Phase 9 — RL wavelet model

First train a simplified policy that selects only the wavelet family. The classifier returns validation macro-F1 as reward. This directly tests the professor-provided RL wavelet idea on PCG.

Phase 10 — Full RL selection

Expand the action to wavelet, level, band, temporal window, and fusion configuration. Use PPO with a discrete multi-categorical action space.

Phase 11 — Adaptive fusion

Process selected representations through temporal, spectral, and wavelet branches. Calculate attention weights and fuse features using $F_{\text{fused}} = \alpha_T F_T + \alpha_S F_S + \alpha_W F_W$.

Phase 12 — Reward and policy update

Use reward = 0.40 Delta macro-F1 + 0.25 Delta sensitivity + 0.20 Delta specificity + 0.10 robustness – 0.05 complexity. Start with macro-F1 only if the full reward is unstable.

Phase 13 — Fine-tuning

Freeze the policy, fine-tune the feature branches and fusion network, then jointly fine-tune with a smaller policy learning rate.

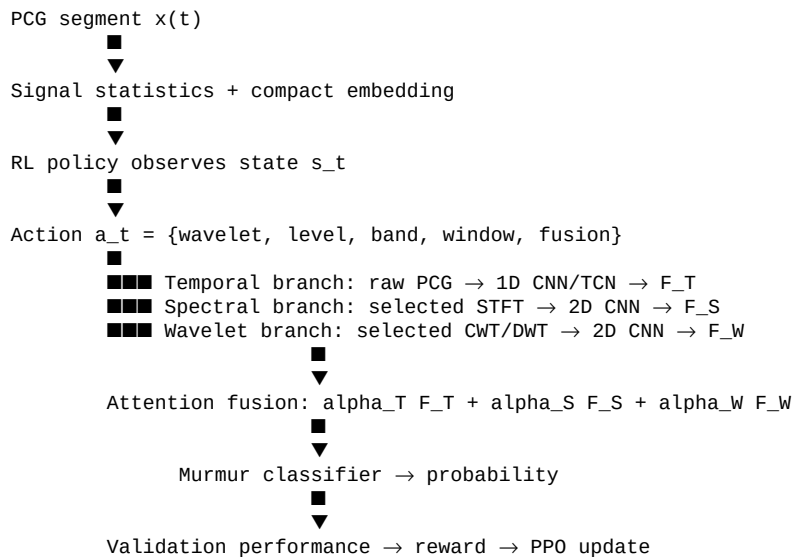
Phase 14 — Final evaluation

Evaluate the frozen model on the patient-independent CirCor test set. Report macro-F1, sensitivity, specificity, balanced accuracy, AUROC, AUPRC, parameter count, and inference time.

Phase 15 — Ablation and explainability

Remove RL, band selection, window selection, wavelet branch, adaptive fusion, robustness reward, and complexity penalty in separate experiments. Visualize selected wavelets, bands, windows, attention weights, and representative PCG regions.

5. Model architecture



6. RL definition

State: $s_t = [\text{SNR, energy, variance, spectral centroid, spectral bandwidth, zero-crossing rate, duration, location encoding, compact CNN embedding}]$.

Action: $a_t = [\text{wavelet family, decomposition level, frequency band, temporal-window size, fusion mode}]$.

Reward: initially $r_t = \text{macro-F1}_t - \text{macro-F1}_{(t-1)}$. After stable training, include sensitivity, specificity, robustness, and computational cost.

Recommended first RL algorithm: PPO. Keep the action space discrete and finite at first; continuous fusion weights can be added only after the discrete version works.

7. Training order

Step	Model	Purpose
1	Raw waveform CNN	Confirm labels and preprocessing work.
2	Fixed STFT CNN	Measure spectral-only performance.
3	Fixed db6 wavelet CNN	Measure wavelet-only performance.
4	Fixed temporal-spectral fusion	Establish a fusion baseline.
5	RL wavelet-only model	Test adaptive wavelet selection.
6	Full RL model	Select wavelet, level, band, window, and fusion.
7	Frozen-policy fine-tuning	Improve final classifier without unstable early joint updates.

8. Evaluation and required tables

Use the primary task murmur present versus murmur absent. Exclude unknown murmur labels from the supervised primary experiment or report them separately. After that, repeat the pipeline for normal versus abnormal outcome if time permits.

Model	Macro-F1	Sensitivity	Specificity	AUROC	AUPRC
Raw CNN	—	—	—	—	—
Fixed STFT CNN	—	—	—	—	—
Fixed wavelet CNN	—	—	—	—	—
Fixed fusion	—	—	—	—	—
RL wavelet-only	—	—	—	—	—
Proposed full RL fusion	—	—	—	—	—

9. Ablation studies

Run at least these comparisons: fixed wavelet versus RL wavelet; fixed band versus RL band; fixed window versus adaptive window; concatenation versus attention fusion; accuracy-only reward versus multi-objective reward; with and without the wavelet branch; and clean versus noisy test signals.

10. Practical first implementation

Do not begin with the complete joint RL system. Implement the following minimum viable sequence: (1) CirCor metadata parser, (2) patient split, (3) preprocessing, (4) fixed temporal-spectral classifier, (5) RL wavelet-only selection, and (6) full joint action. This creates usable results at every stage and prevents the project from failing because the largest RL action space is unstable.

11. Final novelty statement

Unlike conventional PCG classifiers that use fixed wavelets, fixed frequency bands, fixed temporal windows, and predetermined fusion rules, the proposed method uses reinforcement learning to select a signal-specific representation configuration. The agent jointly selects the wavelet family, decomposition level, frequency band, temporal-window size, and fusion strategy, while the reward combines diagnostic performance, robustness to noise, and computational efficiency.

12. Deliverables for the professor

Submit the following items: a dataset description, exact download/version statement, preprocessing flowchart, baseline results, RL action-space table, reward definition, final architecture diagram, ablation table, robustness analysis, selected-action visualizations, and a reproducible README containing the command used to download CirCor v1.0.3.

Dataset citation note

Cite the CirCor DigiScope dataset paper and the PhysioNet dataset page in the final report. State clearly that all experiments use CirCor v1.0.3 only. The PhysioNet Challenge documentation describes the data files and download process, and the dataset page reports the available recording collection. [web:9][web:26]